

## Adhil Hamdan

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### SUMMARY:

Electronics & Communication Engineering graduate and Associate VLSI Engineer with hands-on exposure in Design for Testability at SanDisk, covering DFT RTL development, wrapper generation, and MBIST implementation, with Physical Design exposure at Maven Silicon. Springer-indexed published researcher in image processing and cryptography, with hardware project experience spanning embedded systems, sensor interfacing, and algorithm development. Passionate about Silicon Lifecycle Management, DFT for low-power embedded SoCs, and hardware-software co-design seeking to leverage DFT expertise and a strong hardware foundation to solve complex silicon challenges.

### RESEARCH INTEREST:

- Silicon Lifecycle Management (SLM)
- Design for Testability in Low-Power Embedded SoCs for IoT Applications
- Hardware-Software Codesign

### EDUCATION:

#### **Bachelors in Electronics and Communication Engineering**

**Dec.'21 – Jul.'25**

PSG College of Technology, India (National Institutional Ranking Framework: 67th)

CGPA: 7.93

**Relevant Coursework:** Digital Electronics, Embedded Systems, VLSI, Analog Electronics, Operating Systems, Image Processing

### PUBLICATIONS:

#### **Adaptive Dark Channel Prior Based Image Enhancement for Weather Degraded Images**

**Oct.'25**

Developed an Adaptive Dark Channel Prior (ADCP) image enhancement algorithm for autonomous vehicles to reduce haze and halation, using Otsu's thresholding and edge-preserving filtering to improve image clarity, noise reduction, and dark-region detail visibility.

**Link:** [https://link.springer.com/chapter/10.1007/978-3-031-99939-0\\_24](https://link.springer.com/chapter/10.1007/978-3-031-99939-0_24)

### PROJECTS:

#### **1×3 Router Physical Design:**

**Sep'25. – Dec'25**

- Executed complete Physical Design flow for a 1×3 router using Synopsys Fusion Compiler, covering floorplanning, placement, Clock Tree Synthesis, and detailed routing with optimized cell utilization and congestion management.
- Performed timing closure using Static Timing Analysis (STA) to successfully resolve setup and hold violations, ensuring robust clock distribution with minimized skew and insertion delay.

#### **Dual-Layer Image Security Using Chaotic Maps**

**Dec.'24 – Apr.'25**

- Designed and developed an end-to-end dual-layer image encryption framework in MATLAB utilizing chaotic maps, implementing pseudo-random number-based cryptographic algorithms for secure digital image transmission.
- Validated system performance through rigorous testing, successfully demonstrating enhanced resistance to statistical and differential attacks compared to baseline methods.

#### **Adaptive DCP-Based Image Enhancement (MATLAB):**

**Jun.'24 – Nov.'24**

- Developed an Adaptive Dark Channel Prior (ADCP) framework integrated with fuzzy logic in MATLAB to enhance visibility and reduce atmospheric haze for autonomous vehicle vision systems, optimizing transmission map estimation and atmospheric light refinement to improve contrast and eliminate haze.
- Co-authored and published findings in a Springer journal after presenting at the ICSSIC conference.

#### **Geofencing-Based Speed Limiting System:**

**Jan.'23 – May'23**

- Designed and prototyped an embedded GPS-based speed control system implementing geofencing logic using the Haversine formula in microcontroller firmware, integrating GPS module, Hall Effect sensor, and servo motor for real-time speed restriction within critical zones.
- Achieved high-accuracy zone detection and real-time speed throttling, successfully demonstrating an automated audio/visual alert and control mechanism for enhanced road safety..

#### **Stroke Care 360:**

**Aug.'23 – May'24**

- Designed and engineered an IoT smart pillbox module using a microcontroller, Real-Time Clock (RTC), and DFPlayer Mini to provide scheduled auditory and visual medication alerts, as part of an integrated

assistive healthcare ecosystem for post-stroke rehabilitation.

- Implemented Photoplethysmography (PPG) signal processing algorithms to extract and monitor cardiovascular health metrics in real time, Presented the work at Coimbatore Institute of Technology.

#### EXPERIENCE:

##### **Associate Engineer VLSI**

**Feb.'26 – Present**

*Krivya semicon, Bengaluru, India*

- Involved in DFT workflows at SanDisk through Krivya as a service partner, contributing to IP wrapper generation, RTL creation, IJTAG concepts, padmuxing, MBIST insertion, and ATPG execution.
- Utilized industry-standard EDA tools including Tessent and Spyglass for DFT implementation and verification.

##### **Physical Design Trainee**

**Aug '25 – Feb '26**

*MavenSilicon, Bengaluru, India*

- Completed Physical Design training program covering end-to-end ASIC PD flow including floorplanning, placement, clock tree synthesis, routing, and static timing analysis using industry-standard EDA tools.

##### **Operations Associate Intern**

**Jan'25 – Mar.'25**

*Ninjacart, Bengaluru, India*

- Optimized supply chain operations, inventory management, and process workflows by analyzing large-scale operational datasets to improve fulfillment efficiency and system reliability.
- Collaborated with Product, Analytics, and Operations teams, applying data-driven decision-making to identify bottlenecks and improve performance metrics.

##### **Internship**

*Bright Burnishing Tools Pvt. Ltd. Coimbatore, India*

**May'– Jul.'24**

- Developed a real-time force monitoring solution for industrial burnishing tools using ESP32, integrating load sensors and signal conditioning circuits for accurate force measurement and microcontroller-based data acquisition.
- Calibrated sensors against mechanical standards and analysed force-quality correlation to improve surface finish and tool performance.

##### **Trainee**

*ISRO Propulsion Complex, Mahendragiri, India*

**Dec.'23**

- Gained hands-on experience in field instrumentation, flight instrumentation, hardware, control systems, computer networking, measurement and data acquisition, and command systems.

#### TECHNICAL SKILLS:

- **Programming Languages:** C, python, **tcl**, **MATLAB**, **Verilog**
- **VLSI Physical Design :** Floorplanning, Placement, Clock Tree Synthesis (CTS), Routing, Static Timing Analysis (STA), Physical Verification, Timing Closure, Design for Testability (DFT)
- **EDA & Design Tools:** ModelSim, Intel Q uartus Prime, Fusion Compiler, PrimeTime, Cadence EDA, Synopsys Formality, Tessent, Spyglass
- **Laboratory & Hardware Skills:** Sensor interfacing, signal conditioning, measurement and data acquisition, embedded system prototyping, hardware testing, calibration, and instrumentation, DFT.

#### PROFESSIONAL ACTIVITIES:

- Founder and General Secretary of Green Kanyakumari Foundation, leading environmental and community initiatives. **Oct '22 – Present**
- Served as Member, Publicity Head, and Secretary of IETE Student Chapter, PSG College of Technology, contributing to planning and execution of technical events and student activities **2022 – 2025**
- 1st Place, Nexus, IEEE ATOM 2024, engineering an innovative solution in a technical competition.

**Mar '24**

- Danfoss Innovator Awards (Round 1) Developed a solution for minimizing electricity theft.

**Oct '23**

#### **CONCLUSION:**

Electronics & Communication Engineering graduate and Associate VLSI Engineer with hands-on industry exposure in Design for Testability at SanDisk, contributing to IP wrapper generation, RTL creation, IJTAG concepts, padmuxing, MBIST insertion, and ATPG execution using Tessent and Spyglass, with Physical Design exposure at Maven Silicon. Published researcher with Springer-indexed papers in image processing and cryptography, with strong hardware-software co-design skills spanning embedded systems, sensor interfacing, and algorithm development. With a demonstrated ability to lead and contribute across diverse technical and professional environments, I bring both technical depth and research maturity to drive impactful contributions in advanced VLSI engineering roles.